

Total No. of Questions : 4]

SEAT No. :

P5363

[Total No. of Pages : 1

[6188]-318

B.E. (Computer Engineering) (Honours) (Insem)

MACHINE LEARNING

**Artificial Intelligence and Machine Learning
(2019 Pattern) (Semester - VII) (410301)**

Time : 1 Hour]

[Max. Marks : 30

Instructions to the candidates:

- 1) Answer Q.1 or Q.2, Q.3 or Q.4.
- 2) Assume suitable data, if necessary.
- 3) Neat diagrams must be drawn wherever necessary.

- Q1)** a) Define Machine Learning and state applications of Machine Learning in Image Recognition. [5]
b) Explain the concept of Probably Approximately correct (PAC) Learning in detail. [5]
c) Compare between classification and Regression. [5]

OR

- Q2)** a) What are the advantages and disadvantages of Dimensionality Reduction? [5]
b) Explain Principal Component Analysis in detail. [5]
c) How does Linear Discriminant Analysis Works? [5]

- Q3)** a) Explain how does Logistic Regression works with an example. [7]
b) Write a short note on [8]
i) Ridge Regression
ii) Lasso Regression

OR

- Q4)** a) Differentiate between subset selection and shrinkage methods. [5]
b) What is Least Square Method? Explain. [5]
c) Explain L1 Regularized Logistic Regression in detail. [5]

